

Keyboard Design Notes

Design and Pin Configuration Research

Vahin

July 2026

Contents

1. Initial Design	3
1.1. Design Changes	3
1.2. Edit V2	4
1.3. Links	4
2. Schematic	4
3. New Day, New Keyboard	6
4. Been 3 hours. Still doing traces	7
5. Routing Sucks.	7
6. New Day, More Traces!!!!	8
7. Been another 3 hours, FINALLY PCB SHOULD BE FINISHED!!	9
8. Holy fuck I didn't realise I could hate something more than routing traces	10
9. My Stabiliser Holes are Wrong.	10
10. YIPPEE ITS DONE!	11

1. Initial Design

Note

This section (ie 1. Initial Design) took about 2 hours for research, writing and drawing of the design. on 21/07

Note

The template used in this report is one I made for a school project.

The initial design was based on a drawing of the keyboard, shown in Figure 1. Since actual keyboards are staggered, this was done to make the schematic much easier to create.

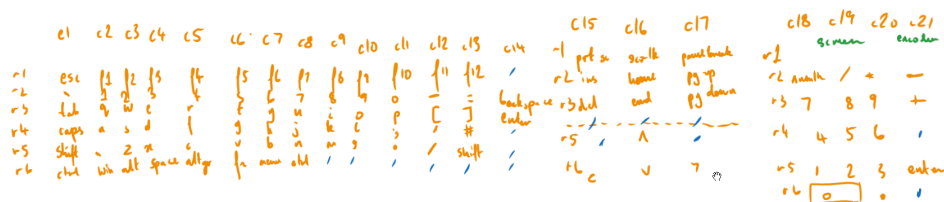


Figure 1: Initial keyboard design drawing.

The design contains a total of 11 empty keys:

1. 3 from r6 c11-13
2. 3 from r4 c14-16
3. 3 from r1 c17-c19
4. 2 from r5 c14 and 16

In total, it is a 19 x 6 matrix. Although this could be changed to an 18x6 matrix since I use 103 keys and the 18x6 can support 108 keys, while 19x6 supports 114 keys, I have not done this purely because its easier to make if I don't need to move a lot of keys around.

1.1. Design Changes

The design changed slightly from the initial version since the rotary encoder takes 2 GPIO pins (if I ignore the button). The earlier design meant I would hit 30 pins, since I assumed a potentiometer was being used rather than a rotary encoder.

As such, I moved some keys around to create an 18x6 matrix, with only 5 empty slots.

The new GPIO usage is:

1. 18 GPIO pins for the matrix rows and columns
2. 2 GPIO pins for the rotary encoder
3. 1 GPIO pin for the addressable LEDs
4. 2 GPIO pins for the OLED using I2C

This gives a total of $24 + 2 + 1 + 2 = 29$ GPIO pins, which is all the microcontroller has available.

In this instance, I have NOT used the pin that the encoder gives when pressed, since that is not something I care about.

Technically the OLED screen has 4 pins, but 1 is GND and one is VCC, so those do not require GPIO. Only the I2C clock and communication pins require GPIO.

1.2. Edit V2

The Orpheus Pico only has 26 pins available, because GP23 is used for the user button, GP24 is used for neopixel, and GP25 is used for the User LED. As such, I need to use 2 MCP23017 modules, which each provide 16 GPIO pins. This means everything can go on those modules, and I can share the I2C pins with the OLED display, meaning fewer pins are needed. This means I have gone back to using more columns for the keys so wiring is simpler. In the end, I have a 21x6 matrix, and 3 rotary encoder pins on the expander, meaning I need 2 GPIO pins for I2C for the OLED display and the expanders, 1 GPIO pin for the LED data, and perhaps 1 more pin for an interrupt on the expander.

1.3. Links

The following links are useful for when the BOM will be made.

1. MCP23017: <https://www.aliexpress.com/item/1005009506514213.html>
2. LED SK6812MINI-E: https://www.lcsc.com/product-detail/C5149201.html?s_z=n_SK6812%2520MINI-E
3. 0.91" OLED: <https://www.aliexpress.com/item/32836691660.html>
4. 100nF Capacitors: <https://www.aliexpress.com/item/1005007470747384.html>
5. 1N4148W OSD-123 Diode SMD-F: <https://www.aliexpress.com/item/1005006323468521.html>

2. Schematic

Note

I spent roughly 2.5 hours on this section on 21/07

I made the schematic for the keyboard with everything wired up. I still have yet to add LEDs for each key.

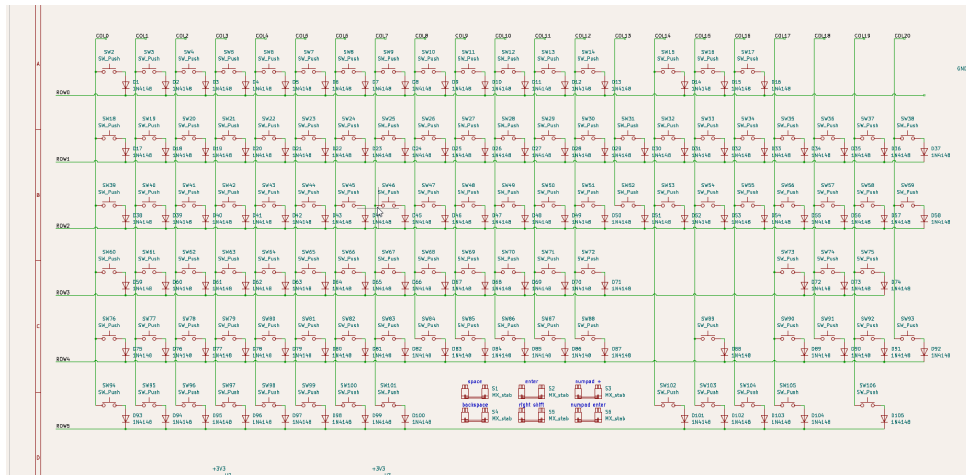


Figure 2: Schematic of the keys

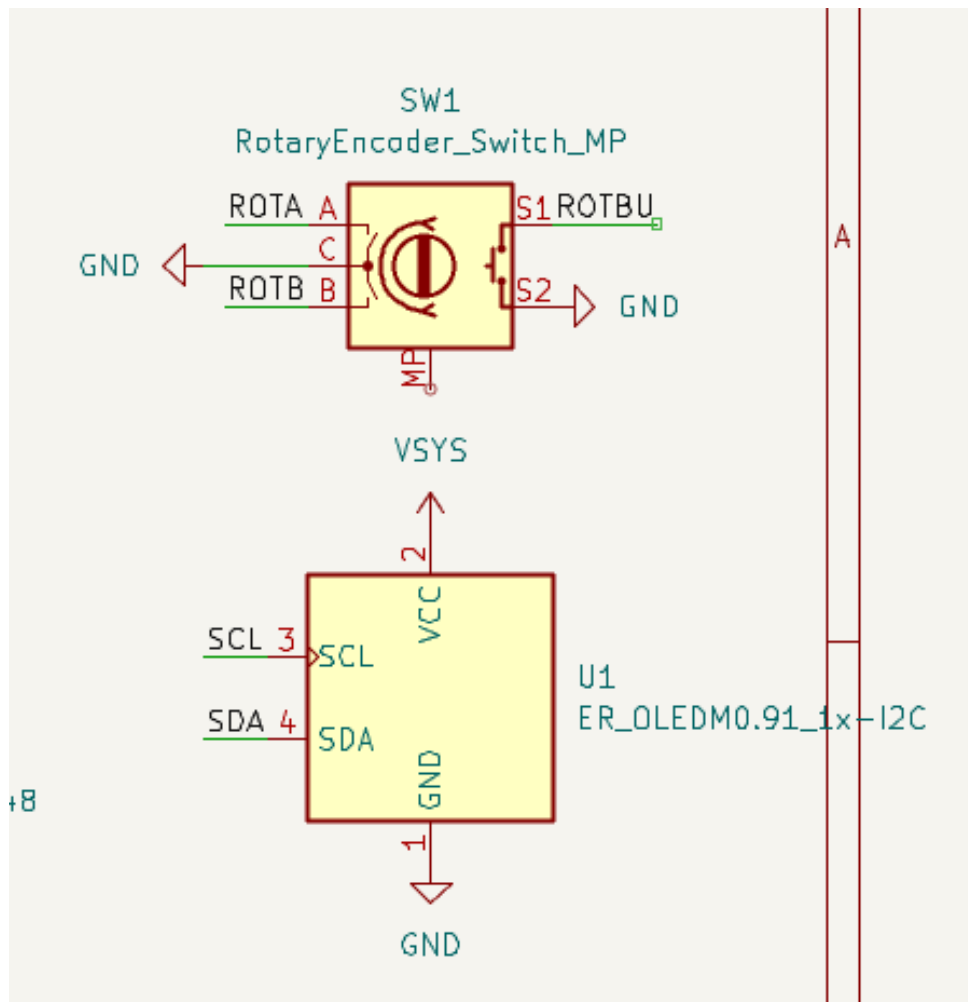


Figure 3: Schematic of the Rotary Encoder and Screen

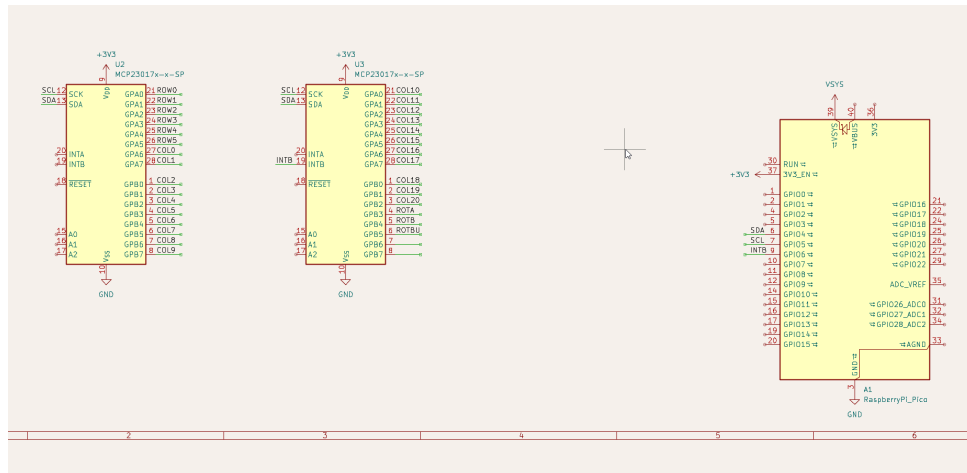


Figure 4: Schematic of the Pi Pico and Expansion Boards

3. New Day, New Keyboard

Originally, I was gonna make this like a proper report. Now, its just my ramblings.

Note

Spent 4 hours redesigning and its not done :(

So, I added LEDs, in this BIG MASSIVE block that looks really ugly. It also turns out that I had one extra extender that I just didn't need, so that was removed. Finally, LEDS ARE AN ABSOLUTE PAIN DON'T GET ME STARTED. FIRST THEY NEED A RESISTOR ON THE DATA IN. SECOND THE SCHEMATIC ALONE TAKES UP 7 LIGHT YEARS OF SPACE AND THIRD WHY ARE THEY SO ANNOYING TO PLACE ON THE PCB ITSELF. THEY NEED TO BE EXACTLY 5.08mm BELOW THE BOARD AND BACK AND THE TRACES ARE A PAIN.

sorry got slightly annoyed.

I moved to adding capacitors to each LED without taking a picture. there should be a commit with the leds done (without capacitors) but I didn't take a pic of it.

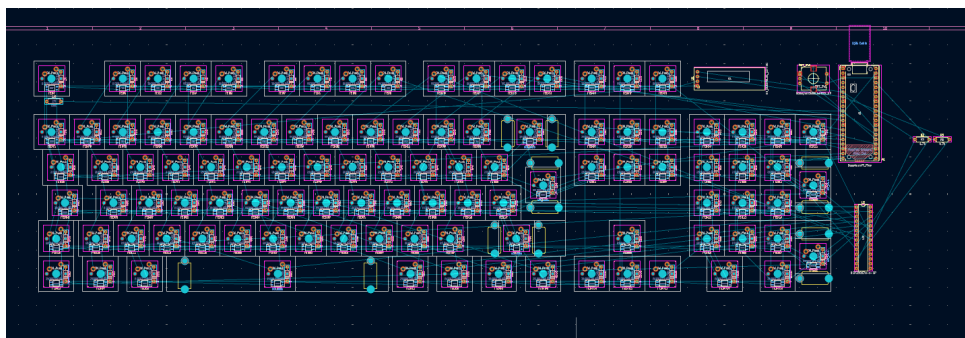


Figure 5: Current PCB design without many traces and without hotswappable footprints

I now plan to add capacitors to the LEDs and move to hotswappable switches.

4. Been 3 hours. Still doing traces

Note

In case the title wasn't clear enough, its been another 3 hours (took an hour off for dinner).

I added capacitors to all of my LEDs, and changed the footprint of every switch to a hotswappable one. Now, I'm currently doing traces, which is no where near finished.



Figure 6: Current PCB design with hotswappable footprints and some traces. This is the bad design with loads of vias.

5. Routing Sucks.

Note

Been another 2 hours, basically been grinding since the last commit.

Routed all of the rows and columns, restarted halfway through since it was really messy. Its a lot better now though. Only have 507 traces left. Apparently, too many vias can disrupt the signal for my LEDs which is also another reason I had to restart, but so far, I haven't used a single via at all. (unlike before)

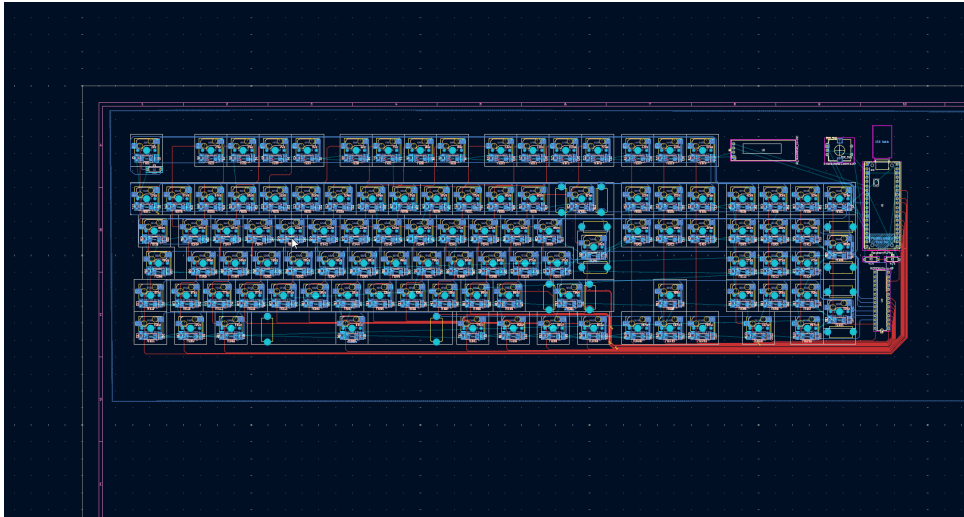


Figure 7: Current PCB design with hotswappable footprints and some more traces

I still need to finish the LED data connections, and improve some of the traces to not be right next to pads, to avoid bridging when actually soldering it. Finally, I still need to do the LED power connections, the OLED connections and the Rotary Encoder connections.

6. New Day, More Traces!!!!

Note

Been 2 hours today

So far today, I have just been moving a lot of my traces to be a little further from the pads so I don't accidentally bridge them when soldering.

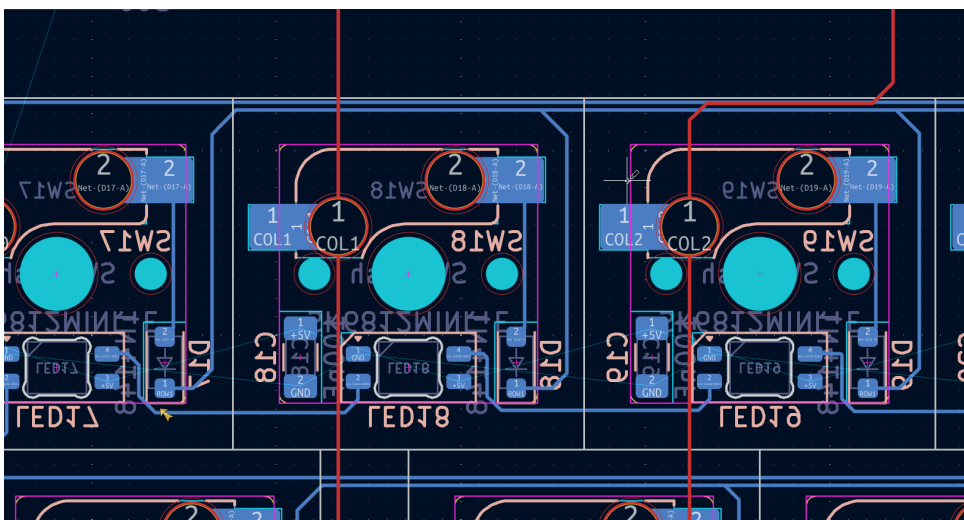


Figure 8: Traces moved slightly

The ROW traces and the LED DOUT traces are mainly the ones that I moved.

Note

Been another hour and a half, finished the LED DOUT traces, and did some of the 5V traces.

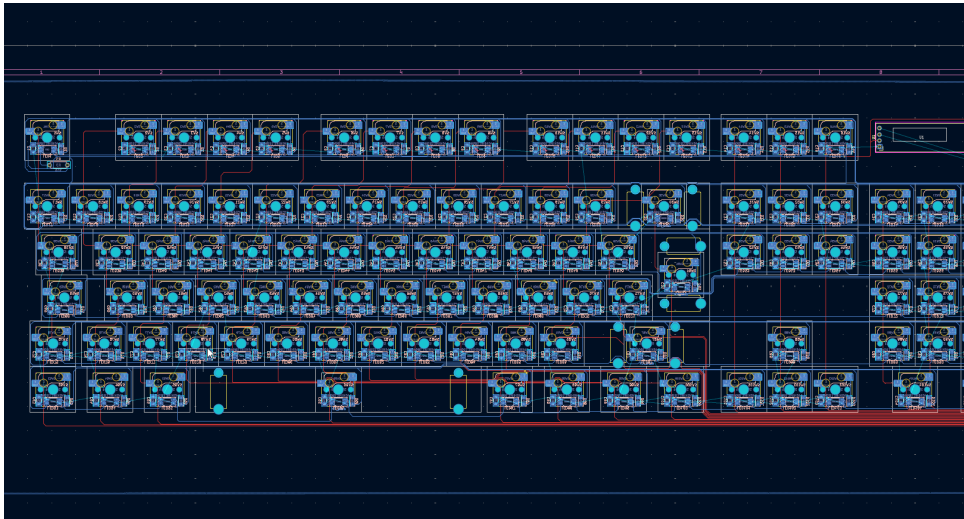


Figure 9: Routing with all LED Data Done.

7. Been another 3 hours, FINALLY PCB SHOULD BE FINISHED!!

Note

Section title, its been 3 hours. The PCB is done!

I finished the PCB!!!!!!

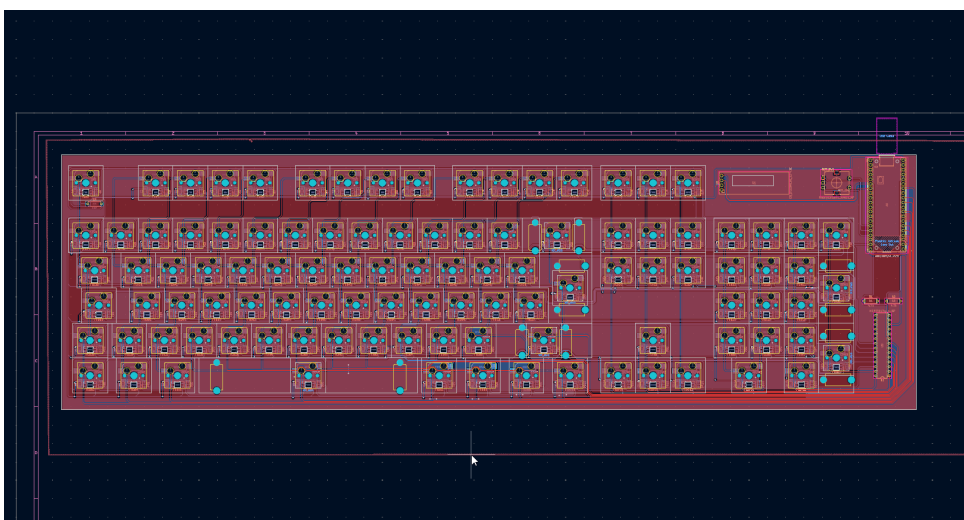


Figure 10: ITS ALL DONE!!!!!!

8. Holy fuck I didn't realise I could hate something more than routing traces

Note

Been 2 hours. Minimal progress. ALL BECAUSE OF FUSION 360.

So first, I export my PCB as a .step file right. Try to import that into fusion. It should work? WRONG. IT WON'T ACCEPT A FILE WITH THE NAME ".step". THAT TOOK A SOLID 40 MINUTES TO DEBUG I KEPT CHANGING EXPORT SETTINGS AND NOTHING WORKED.

Then, since I used the Kalix hotswappable sockets, I needed to add Switches to each hole for me to be able to line it up properly. I get through like half of the keys. THEN I REALISE THE MODEL I USED WAS WRONG AND CAME WITH AN LED IN THE DAMN SWITCH THAT WAS GOING THROUGH MY DIODE ON THE BOTTOM. THEN FUSION WON'T LET ME DELETE THE LED SO I NEED TO GO FIND A DIFFERENT MODEL. AND BECAUSE YOU CAN'T COPY AND MULTI PASTE WITH POINT TO POINT IT TAKES SO LONG TO PLACE EACH KEY.

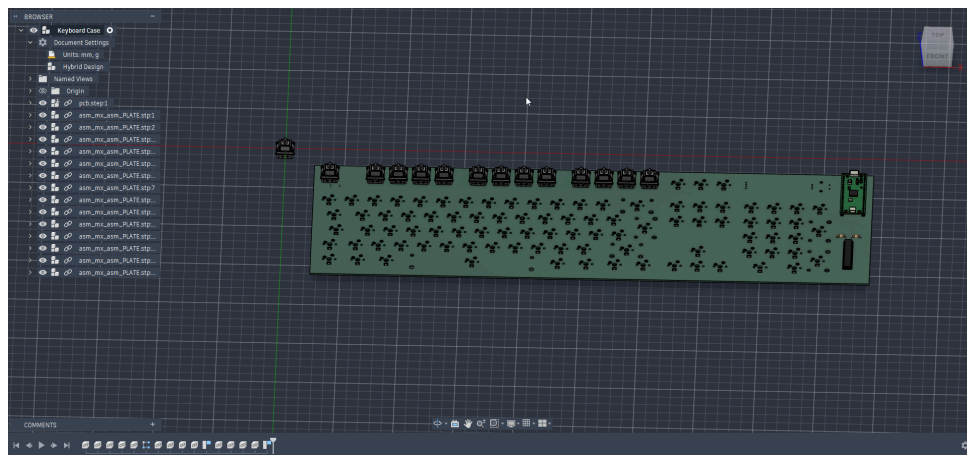


Figure 11: The Design in Fusion for now

9. My Stabiliser Holes are Wrong.

Note

This took about an hour.

So, I finished adding my switches to the keyboard, some are ever so slightly off but it doesn't matter too much, and then added a plate and extruded it. However, the keyboard plate I generated from <https://kbplate.ai03.com/> to match the keyboard layout I used has a big stabiliser hole. EXCEPT THE HOLES THE MX_STAB ADDS DON'T FIT IN THAT HOLE. It is now 3:45am, and I will not stand for this. We'll see it in the morning.

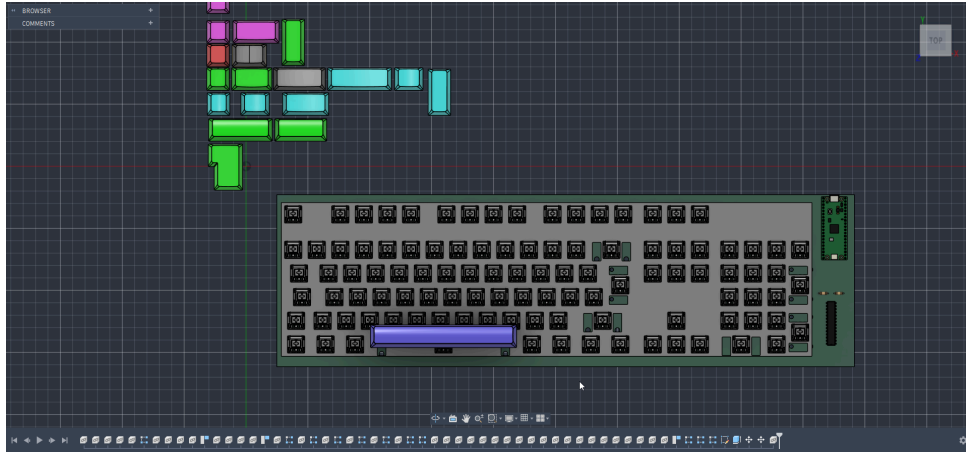


Figure 12: The Design in Fusion for now v2

10. YIPPEE ITS DONE!

Note

Been 4 hours. Shouldn't have taken this long but I kept needing to change some small things here and there and I'm not very used to fusion. Probably would have taken a pro only around 2/3.



Figure 13: The Final CAD.

I'M DONE! I extended the plate to cover the bottom frame parts as well, added mounting holes and such for the case, and filleted my corners. I also made holes in the plate for the OLED screen and rotary encoder. This should be it.